MODULE-4

EMBEDDED SYSTEMS

Smart watches → combine sensors, processors, & wireless connectivity to track health data, display notification & interact with smart watch

Embedded System:

An embedded system is an electronic/electro-mechanical s/m designed to perform a specific function & is a combination of both hardware & software.

Embedded systems v/s General computing system

Embedded System	General computing system
1) Combination of special purpose hardware & embedded OS for executing a specific set of applications	1) Combination of generic hardware & a General purpose OS for executing a variety of applications.
2) May or may not contain an OS for functioning.	2) Contain a General purpose OS (GPOS).
3) The firmware of the embedded system is pre-programmed and it is non-alterable by end-user	3) Applications are alterable by user.
4) Applications specific requirements are the key deciding factors	4) Performance is the key deciding factor on the selection of the s/m.
5) Execution behaviour is deterministic for certain types of embedded systems like 'Hard Real Time' s/m	5) Need not be deterministic in execution behaviour.

Classification of Embedded Systems

Organizing & optimizing a

1) Based on generation μP & μC are different ways of computing s/m
μP only consists CPU, μC has memory, CPU & I/O

» First Generation: The early embedded system were built around 8-bit microprocessors like 8085 & Z80, and 4-bit microcontrollers
→ Simple in hardware circuits with firmware built/developed in Assembly code. → low level programming language that allows developers to write program that run directly on a CPU
→ Ex: Digital telephone keypads, stepper motor control units, etc.

» Second Generation: These are embedded systems built around 16-bit microprocessors & 8 or 16-bit microcontrollers.
→ The instruction set for second generation processors/controllers were much more complex & powerful than the first generation processors
→ Ex: Data acquisition system, SCADA s/m, etc

>> Third Generation : These are embedded systems built around 32-bit processors & 16-bit microcontrollers
Ex: Industrial process control, networking, etc.

>> Fourth Generation : The fourth generation embedded systems are making use of high performance real time embedded operating system for their functioning.
→ Use of advanced microcontrollers, FPGAs & SOCs (System on chips)
→ High performance with energy efficiency.
Ex: Smartphones, smart home devices.

OS → manages computer hardware & s/w resources & provides common services

2) Based on Complexity and performance

>> Small scale Embedded systems : These are usually built around low performance & low cost 8 or 16-bit μp/μc
→ It may or may not contain an operating s/m for its functioning
Ex: Electronic toy

>> Medium scale embedded systems : These are usually buit around medium performance & low cost 16 or 32-bit μp/μc or DSP.
Ex: Decoding / Encoding of media, crypl

>> Large Scale embedded systems : Buit around high performance 32 or 64-bit μc/μp.
Ex: Decoding / Encoding of media.

✱ Major Application areas of Embedded system

1) Consumer electronics
   Ex: Camcorders, cameras, etc
2) Household appliances
   Ex: TV, DVD players, fridge, etc
3) Home automation & security systems
   Ex: Air conditioners, fire alarms, etc
4) Automotive industry
   Ex: Anti-lock breaking system (ABS), engine control
5) Telecom
   Ex: Cellular phones, telephone switches, etc
6) Computer peripherals
   Ex: Printers, scanners, etc
7) Computer networking system
   Ex: Network routers, switches, etc.

8) Healthcare
Ex: EEG, ECG machine, etc.

9) Measurement & Instrumentation
Ex: Digital multi meters, CROs, etc

10) Banking & Retail
Ex: Automatic teller machines (ATM) & currency counters

11) Card Readers
Ex: Barcode, smart card readers, etc.

⊛ Core of the Embedded system

The core of the embedded system falls into any one of the following categories

1) General purpose & domain specific processor
   - → Microprocessor
   - → Microcontroller
   - → Digital signal processor

2) Application specific Integrated circuit (ASIC)
3) Programmable logic devices (PLDs)
4) Commercial off-the-shelf Components (COTS)

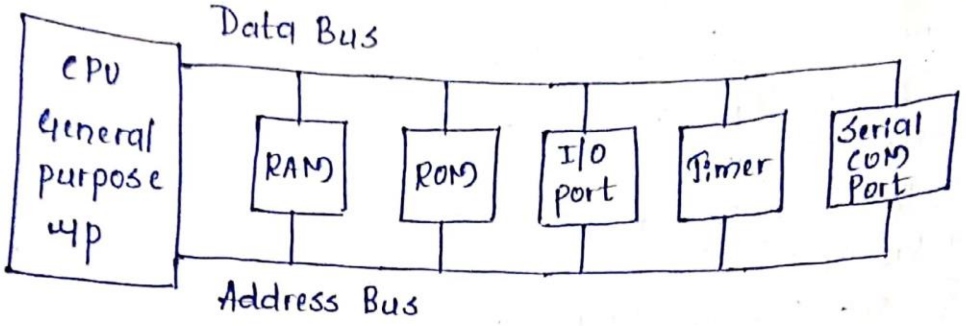
>> Microprocessor v/s Microcontroller

Microprocessor	Microcontroller
1) We need to connect peripherals externally. So it makes circuit bulky.	1) The presence of peripherals such as RAM, ROM, I/O, & timers are In-built. So it is available on a single chip.
2) It increases the overall cost of the s/m high.	2) The overall cost of the system is less.
3) We can connect external memory in range of Mbytes & even Gbytes. But speed is less.	3) The inbuilt finite memory helps to improve the speed of operations.
4) Due to external components, the total power consumption is high. ∴ it is not ideal for the devices running on stored power like batteries.	4) As external components are low, total power consumption is less. So it can be used with devices running on stored power like batteries.
5) Most of the µp do not have power saving feature.	5) Most of the µc offer power saving feature.

RISC VS CISC :-

RISC (Reduced Instruction set Computing)	CISC (Complex Instruction set Computing)
1) Lesser number of Instructions	1) Greater number of Instructions
2) Instruction pipelining and increased execution speed.	2) No instruction pipelining feature.
3) Orthogonal Instruction set.	3) Non-orthogonal Instruction set.
4) operations are performed on registers only, the only memory operations are load and store	4) operations are performed on registers or memory depending on the instructions.
5) A large number of registers are available	5) Limited number of general purpose registers
6) Single, fixed length instruction.	6) Variable length instructions
7) Less silicon usage and pin count	7) More silicon usage
8) with Harvard Architecture	8) Can be Harvard or Von-Neumann Architecture.

→ Microprocessor



Defn: A microprocessor is a silicon chip that functions as a Central Processing Unit (CPU). It performs arithmetic & logical operations. However it is a dependent unit, meaning it does not include I/O ports, RAM, ROM on the chip itself. These components must be connected externally.

- CPU: Executes instructions, performs arithmetic, logical & control operations.
  → Acts as brain of the system, controlling the flow of data between various components.
- Data Bus: Transfers data between the CPU, memory and peripherals.
- Address Bus: Carries the address of the memory location or I/O device that the CPU wants to access.
- RAM (Random Access Memory): Temporary storage for data & instructions that the CPU is currently using.
  → Volatile memory (data is lost when power is off)
- ROM (Read Only Memory): Non-volatile memory that stores permanent instructions / firmware.
- I/O ports: Allow the µp to communicate with external devices like keyboards, displays or storage units.
- Timer: Provides clocking functions for timing operations, like delays / synchronization
- Serial Communication port: Enables communication with external devices via serial protocols like UART or SPI.

UART ↓ Universal Asynchronous Rx Tx
It is a protocol that allows 2 devices to exchange serial data over a single line.

» Microcontroller

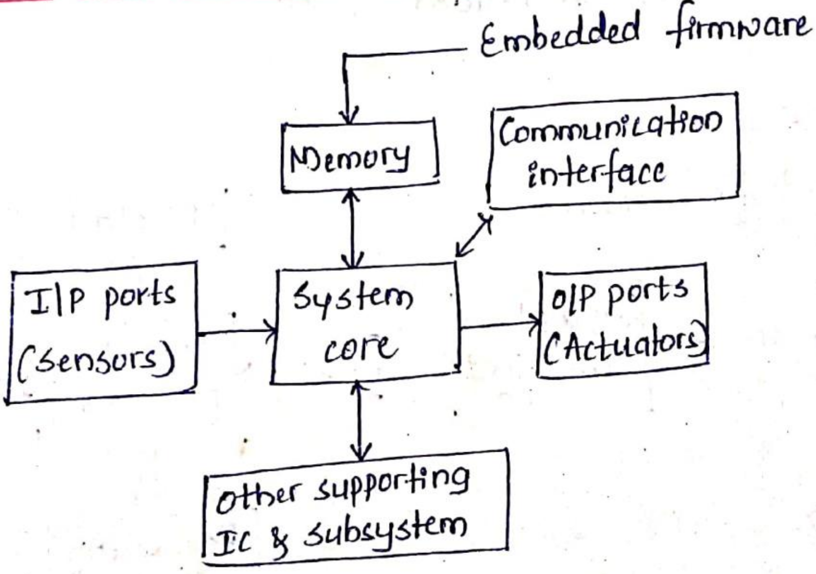
CPU	RAM	ROM
I/O Port	Timer	Serial COM port

← A single chip

Def^n^: It is a highly integrated chip contains CPU, scratch pad RAM, on chip ROM/flash memory, I/O ports & is a self contained unit. It is application oriented & domain specific, it also contains multiple built in I/O ports & lot of power saving features.

Ex. Motorola's 6811, Intel's 8051, Zilog's Z8 & PIC 16X

⊛ Elements of Embedded System



fig(a) Embedded System

⇒ Embedded system contains a system core that is a single chip controller which acts as a master brain of the system. The controller can be microprocessor or a microcontroller or DSP or ASIC.

› Processor used for processing the signal & data to execute desired set of instructions with high speed of operation.

⇒ Embedded s/m's are basically designed to regulate a physical variable or to manipulate the state of some devices by sending some signals to the actuators, in response to the input signal provided by the end user or sensors which are connected to the

(*) Definition of sensors :-

A sensor can be defined as an input device which provides an output w.r.t a specific physical quantity (input).

I/P signal → | Sensor | → O/P signal

→ Convert physical characteristic into electrical signals
→ It takes input from environment
→ It is placed at input part of the system.
→ It is used to measure physical quantity.
→ It gives information to the system about environments.

(*) Classification and selection of sensors.

>> Classifications of Sensor

1. Active and Passive Sensor :- → Based on Power Requirement
   → Active sensors are those which requires an external power signal to generate output
   → Passive sensors on the other hand, do not require any external power signal and directly generates output response.

2. Classification is based on the means of detection used in the Sensor. Some of the means of detection are electric, Biological, chemical, Radioactive, etc.

3. Classification is based on conversion phenomenon i.e., the input & the output. Some of the common conversion phenomenon are photo-electric, Thermoelectric, Electromagnetic, Electrochemical, Thermo optic, etc.

Based on O/P Type

4. Analog and Digital Sensors → Analog sensors produce an analog o/P i.e., a continuous o/P signal w.r.t to the quantity being measured. Digital sensors → It provide discrete or digital data as o/P.

Transducer

→ A transducer is a device that converts one form of energy into another.

→ Transducers convert physical phenomena (like sound, light, temp, etc) into electrical signal or vice versa.

One form of energy → [Transducer] → Another form of energy

→ The process of conversion of one form to another is known as transduction

→ Thus, we can define a transducer as a device that converts non-electrical energy to electrical energy.

→ It consists of a sensor and transduction element connected in series.

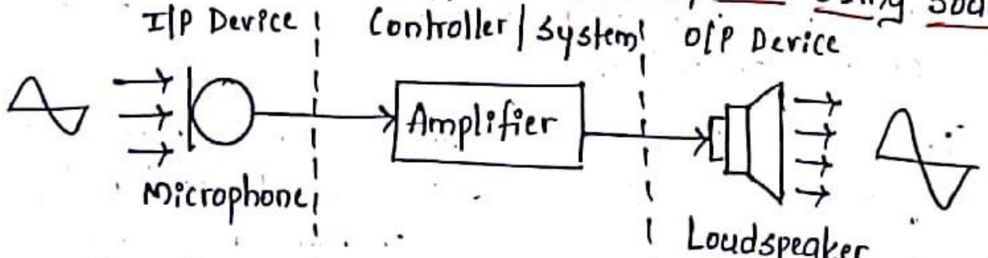
Input Signal → [Sensor] → [Transduction element] → Output Signal

Sensor : Detects a specific physical parameter & converts it into an electrical signal.

Transduction element : Processes the electrical signal from the sensor and converts it into another form of energy or action.

Ex : Microphone, Loudspeaker

Example of a Mobile phone Call System Using sound transducer



Microphone (Sensor / I/P Device)

→ Converts sound waves (acoustic energy) into electrical signals

→ For ex, in a mobile phone, the microphone picks up the user's voice and converts it into an electrical sig that can be processed and transmitted.

Amplifier (Transduction Element)

- Processes & amplifies the electrical signal from the microphone to a level suitable for T_x^n & driving the speaker.
- **Ex**: Amplifier ↑se the strength of the sig to ensure the sound is loud enough when o/p through the speaker

Speaker (o/p Device / Actuator)

- Converts amplified electrical signal back into sound waves
- **Ex**: In a mobile phone, the speaker outputs the received & processed signal, making it audible to the listner.

⊛ Classification of Transducer

Classification of Transducers

- Based on energy source requirement
  - Active
  - Passive
- Based on physical phenomenon
  - Primary
  - Secondary
- Based on its output
  - Analog
  - Digital
- Based on Transduction phenomenon
  - Transducer
  - Inverse T
- Based on construction
  - Mechanica
  - Electrical
  - Optical

1. Based on Energy requirement

a) **Active Transducer** :- These transducers generate electrical output directly from the physical quantity without requiring an external power source.

One form of energy → [Active Transducer] → Electrical signal

Ex: Photovoltaic / solar cell
- Converts light energy directly into electrical energy.

b) **Passive Transducer** :- These transducers require an external power source to operate and produce an output.

One form of energy → [Passive Transducer] → Electrical Signal

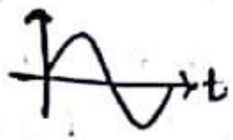
[Power Source] → [Passive Transducer]

Ex: Resistance Thermometer (RTD)
- Measures temprature by changing resistance

3. Based on its Output :

a) Analog Transducer :- Produces a continuous output signal proportional to the measured quantity or Convert the I/P quantity into an analog O/P which is a continuous function of time

Physical Quantity → Analog Transducer → Analog Electrical signal

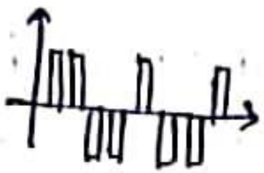


Ex : Thermistor

→ Measures temprature through resistance variation

b) Digital Transducer :- Converts input signal into the output signal in the form of pulses / digital data.

Physical Quantity → Digital Transducer → Digital Electrical signal



Ex : Photo Diode

→ Converts light into electri-cal signals that can be interpreted digitally.

Actuators:

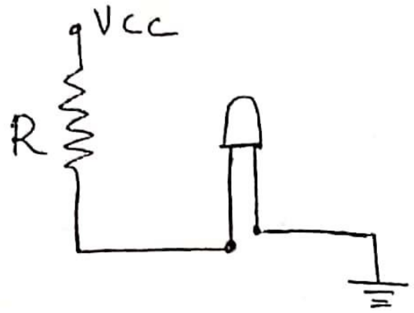
* Actuators is a form of transducer devices which converts mechanical or electrical signals to corresponding physical action such as motion.
* Actuators acts as an output device.

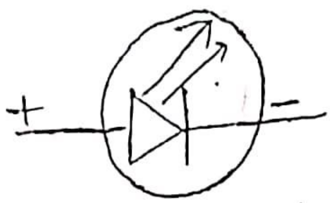
LED: (Light Emitting diode)

* LED is an important output device.
* LED is a p-n junction diode and it contains anode and a cathode.
* For proper functioning of the LED, the anode should be connected to +ve terminal ~~and~~ of the supply voltage and cathode should be connected to -ve terminal of the supply voltage

⇒.

* The ideal LED interfacing circuit is shown in fig.





symbol of LED

* The current flowing through the LED must be limited to a value below the maximum current that it can conduct.
* A resistor is used in series between the power supply and the LED to the current through the LED.

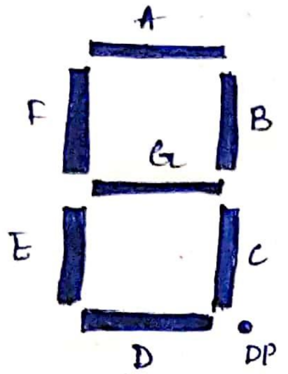
Advantages of LED

*) lower energy consumption
2) longer life time
3) improved physical robousness
4) smaller size
5) Faster switching.

Application of LED

1) Automotive head lamps.
2) traffic signals.
3) general lighting
4) camera flashes
5) media devices
6) LED TVs.

7 segment LED display



→ The 7-segment LED display is an o/p device for displaying alpha numeric characters.

→ It contains 8 light emitting diode (LED) segments arranged in a special form.

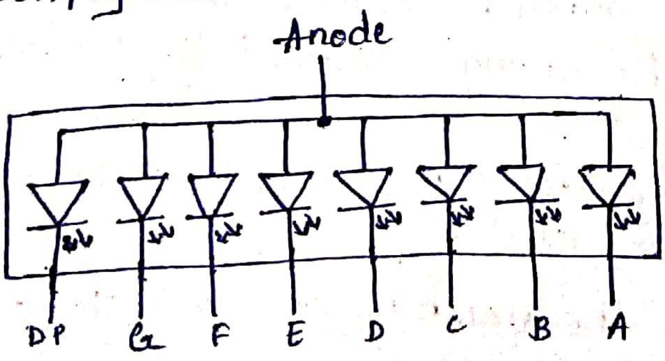
→ Out of the 8 LED segments, 7 are used for displaying alpha numeric characters & 1 is used for representing 'decimal point' (DP) in decimal number display.

→ The LED segments are named A to G & the decimal point LED segment named as DP.

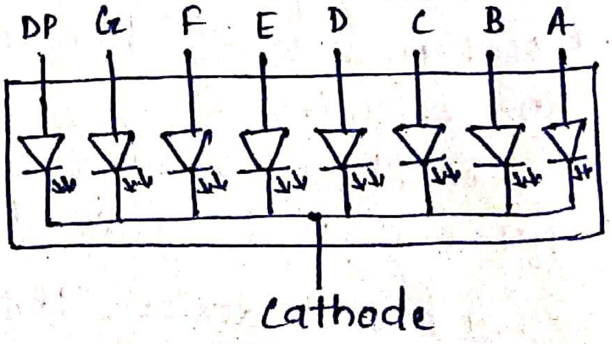
→ The LED segments (A to G) & DP should be lit accordingly to display numbers & characters.
For ex, for displaying the number 4, the segments F, G, B & C are lit.
For displaying the character 'd', the segments G, E, D, C & B are lit.

→ All these 8 LED segments need to be connected to one port of the processor/controller for displaying alpha numeric digits.

→ The 7 segment LED displays are available in two different configurations, namely Common Anode & Common Cathode configuration



Common Anode configuration



Common cathode configuration

→ Based on the configuration of the 7-segment LED unit, the LED segments anode or cathode is connected to the port of the processor/controller in order 'A' segment to the least significant port pin & DP segment to the most significant port pin.

→ In the common anode configuration, the anodes of the 8 segments are connected commonly, whereas in the common cathode configuration, the 8 LED segments share a common cathode line.